

GHK-Cu

Copper Peptide GHK-Cu

SKIN REPAIR

ANTI-AGING

COLLAGEN

HAIR GROWTH

TOPICAL/INJECTABLE

WHAT IS IT?

GHK-Cu (glycyl-L-histidyl-L-lysine:copper) is a naturally occurring human plasma peptide that declines significantly with age. It acts as a master regulator of tissue repair and renewal — promoting collagen synthesis, wound healing, hair follicle regeneration, and gene expression reprogramming toward a more youthful state.

DOSING AT A GLANCE

SC/IM (injectable)	Topical creams	Frequency	Duration
0.5–2 mg	Topical 1–3%	Daily	Ongoing

KEY BENEFITS

- Stimulates collagen and elastin synthesis — improves skin firmness and elasticity
- Reduces fine lines and wrinkles (validated in multiple clinical studies)
- Promotes wound healing and scar remodeling
- Stimulates hair follicle proliferation — reduces hair loss, promotes regrowth
- Anti-inflammatory — reduces pro-inflammatory cytokines
- Antioxidant — chelates copper into protective pathways
- Reprograms aged gene expression toward younger patterns (Pickart research)
- Potential anti-cancer properties via tumor suppressor upregulation
- Neuroprotective — enhances nerve growth factor

HOW IT WORKS

GHK-Cu acts as a biological 'switch' that resets aging gene expression. It upregulates over 31 genes related to tissue repair and downregulates 36 genes associated with inflammatory signaling and cancer progression. It also activates proteasomes (cellular cleanup systems) and promotes angiogenesis in damaged tissue.

PROTOCOLS

- **Topical Anti-Aging:** 1–3% GHK-Cu serum applied 2x daily to face/scalp.

- **Injectable:** 0.5–2 mg SC daily or 3–5x weekly.
- **Hair Loss:** Scalp massage with GHK-Cu serum + topical minoxidil for synergy.
- **Stacks well with:** BPC-157, TB-500 for enhanced healing protocols.

SIDE EFFECTS

- Topical: mild skin redness, irritation (transient, usually at high concentrations)
- Injectable: injection site reactions (mild)
- Generally considered very safe — naturally occurring in human plasma
- Rare: skin discoloration at injection sites

STORAGE

Lyophilized: room temp, away from light. Topical serums: refrigerate to extend shelf life.

■ ***Use copper-peptide compatible skincare — avoid high-pH products (Vitamin C serums, retinoids) simultaneously as they degrade peptide.***

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